Module 04: Cognition in Middle School

Learning Objectives

1. Define **Cognition** within the context of Middle School.
2. Analyze how Cognition interacts with other components of the Active Inference framework.
3. Apply specific constraints of Middle School to the formal definition of Cognition.

Introduction

This module explores **Cognition**. In the **Middle School** curriculum, we approach this topic with a focus on specific applications and theoretical depth appropriate for the audience. Cognition is a critical component of the 8-part Active Inference spine, bridging the gap between Perception and Action.

Key Concepts

1. Cognition as a Markov Blanket Boundary

How does Cognition define the boundary between the agent and the environment?

2. Generative Models of Cognition

What parameters involved in Cognition must be optimized to minimize variational free energy?

3. Active Inference Dynamics

How does the process of Cognition drive the perception-action loop?

Applications

In Middle School, we see Cognition manifest in: * **Specific Example 1**: When a chess program like Stockfish evaluates a position, it builds a tree of possible moves, predicts what the opponent will do in response, simulates several turns ahead, and scores each outcome. That branching simulation is computational cognition -- the program is running a generative model of the game to decide which move leads to the best predicted future. * **Specific Example 2**: When you debug a program, you are doing cognition about code. You read the error message, build a mental model of what the program is doing on each line, predict where the bug might be ("I bet the loop runs one time too many"), and then check. If your prediction is wrong, you update your model and try a new hypothesis. Debugging is literally the prediction-error-update cycle applied to coding.

Conclusion

Understanding Cognition allows us to better model complex adaptive systems. In the next module, we will expand on this foundation.